Loan Loss Provisions: Anticipatory or Reactionary?

A Macro-Information Responsiveness Analysis

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What's a Bank?

A Bank to us is:

- A place to store money securely
- A place to borrow money for big purchases
- A place to invest money for growth

A Bank is a Business:

- The loans you take out are their products
- Banks make money by lending money, and charging for this service in the form of interest
- This type of product is risky however, sometimes people don't pay back their loans

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Imagine this Scenario:

- A bank loans out 1 million dollars to 100 customers
- Each customer is expected to pay back $10,000 with interest
- Because loans are risky, it's an overestimation to say that all 100 customers will pay back their loans

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Loan Loss Provisions (LLP)

- To prepare for potential losses, banks set aside a portion of their profits as an "Allowance for Loan Losses" (ALL)
- Loan loss provisioning is the process of estimating, adjusting and recording this allowance
- It acts as a financial buffer to absorb potential losses from defaulted loans

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Simple Idea:

- If banks expect more losses in the future -> they will increase their LLP
- If risk looks low -> they will decrease their LLP

Loan Losses are Time Sensitive:

- When faced with increasing risk, banks must adjust their LLP to mitigate potential losses
- Banks can try to **anticipate** future losses
- Or they can **react** to losses once they are visible

The Tradeoff:

If Banks Adjust too Early:

- They might over-provision, which can reduce profitability and competitiveness
- Risk may be overestimated, leading to inefficient capital allocation
- It can signal to the market that the bank is in trouble, even if it isn't
- Fear of missing out / being left behind

If Banks Adjust too Late:

- Losses appear suddenly, causing financial instability
- Profits drop sharply, compromising investor confidence
- Exposure to risk increases, potentially threatening the bank's long-term viability
- Fear of not getting out

If Banks had Perfect Foresight:

- They would adjust their LLP perfectly in line with future losses
- They would recognize risk and adjust before these losses occur
- They would anticipate signals ahead of their competition, freeing up captial during good times, and building up buffers during bad times

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The Reality:

- Banks have imperfect information about the future, and must rely on noisy/uncertain signals to guide their decisions
- Acting too early isn't always a safe bet, it has costs
- LLP only adjusts quarterly, meaning a natural delay and potential to miss signals

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Research Question

How do banks adjust loan loss provisions over time?

- Do banks **anticipate** economic downturns?
- Do they **react after conditions worsen?**
- Or do they adjust **at the same time** as the economy changes?

Secondary Question

- What role can **data science** play in improving how banks recognize risk and set provisions?

How Do We Study This?

> We want to understand **when** LLP responds to economic changes via a multi-signal, time-series analysis

- Not just whether economic changes variables are related to LLP
- But **how that relationship unfolds over time**

What Do We Mean by Economic Changes?

Loan repayment is a conflux of economic factors…

We use a small set of indicators to capture different limitations on borrowers' ability to repay, and include metrics that signal credit risk/financial stress

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What Signals Do We Use?

We use indicators capturing different dimensions of risk:

Unemployment Rate

- Measures the share of the labor force without a job

Why it matters

- Job loss reduces borrowers' ability to repay loans
- Higher unemployment → higher default risk

Captures household financial stress

Industrial Production

- Measures output in manufacturing, mining, and utilities

Why it matters

- Proxy for the greater economy as a whole
- Declining signal suggest slowing economic activity
- Lower output → weaker income and business conditions

Captures real economic activity

Term Spread

- Difference between long-term and short-term interest rates for U.S. Treasury bonds

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Why it matters

- Reflects expectations about future economic conditions
- Inversions often precede/predict recessions

Captures economic expectations and monetary conditions

Credit Spread

- Difference between corporate bond yields and safe government yields

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Why it matters

- Wider spreads indicate higher perceived default risk
- Reacts quickly to changes in financial conditions

Captures market-implied credit risk

National Financial Conditions Index (NFCI)

- Composite index of financial market stress

Why it matters

- Aggregates information from credit, leverage, and risk markets
- Increases during periods of financial instability

Captures system-wide financial conditions

Small Business Optimism Index

- Survey-based measure of business expectations

Why it matters

- Reflects expectations about hiring, sales, and growth
- Provides a forward-looking view of economic conditions

Captures business sentiment and expectations

From Economic Signals to Analysis

- We now have a set of indicators capturing different types of risk

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The key question becomes:

How does LLP respond to these signals over time?

The Core Challenge

- Economic variables and LLP are both **time series**
- Relationships are not instantaneous

→ Responses may occur:

- Immediately
- With delay
- Or even in advance

→ We need a framework to capture **timing**

Our Approach

We use a simple - 3 step - time-series framework to study:

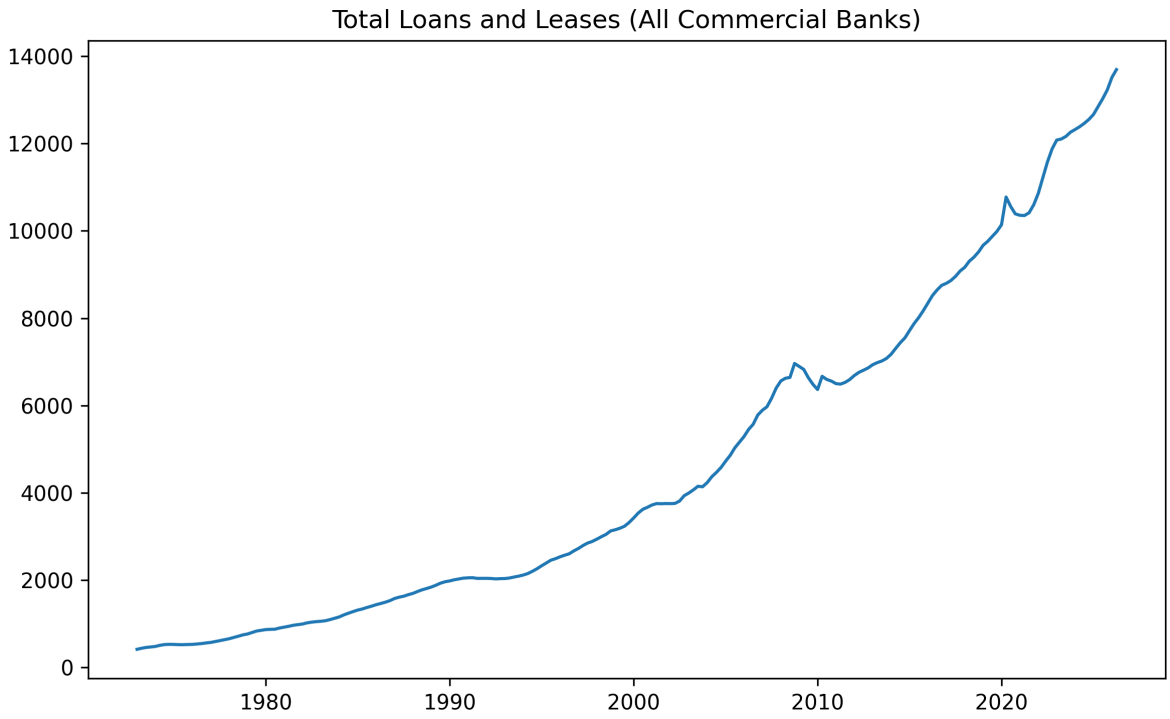
- **When** LLP responds
- **How strongly** it responds
- **How long** responses persist

→ Focus is on **dynamic relationships**, not prediction

Step 1 — Focus on Changes

- We convert all variables into **quarterly changes**

Why?



- Removes long-term trends like bank sector growth
- Makes variables comparable
- Isolates **adjustment behavior** in the face of changing conditions

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Step 2 — Introduce Timing

We compare LLP today to:

- Current values of macro variables
- Past values (**lags**)
- Future values (**leads**)

Leads and Lags

- **Lagged variables**
  → Does LLP respond after changes occur?
- **Contemporaneous variables**
  → Does LLP move at the same time?
- **Lead variables**
  → Does LLP move before changes occur?

→ This allows us to test **anticipation vs reaction**

Step 3 — Regression Framework

We estimate relationships of the form:

- Changes in LLP explained by
  - current signals
  - past signals
  - future signals

→ This captures how LLP evolves **across time**

What Are We Looking For?

Prediciting LLP is not within the scope of this analysis.

We are interested in:

- **Timing of responses**
- **Direction of relationships**
- **Consistency across variables**

→ Do banks recognize risk **early, on time, or late?**

Why This Framework Works

- LLP is a **dynamic adjustment process**
- Economic signals evolve over time

→ This framework allows us to observe:

- Information processing
- Delays in response
- Evidence of anticipation

How regression works as a tool for understanding timing:

- LLP is not set randomly
- It responds to information about the economy

At each point in time:

- We ask:
  → When a macro variable changes, how does LLP respond?

The regression quantifies this relationship

→ The regression helps us trace:

- How information is processed
- How quickly banks react
- Whether they anticipate future risk

Interpreting the Results

Each coefficient tells us:

- **Direction**
  → Does LLP increase or decrease?
- **Timing**
  → Does the response happen before, during, or after?
- **Strength**
  → How large is the response?

Model Choice

- OLS provides a **baseline, interpretable framework**

- Allows us to isolate:
  - Timing effects
  - Direction of relationships
  - Consistency across variables

→ More complex models can be explored later
→ But OLS is ideal for **initial structural understanding**

Why Not More Complex Models?

- Advanced models (e.g., machine learning, VARs):
  - Improve prediction
  - But reduce interpretability

Tradeoff:

- Complex models → better fit
- Simple models → clearer insight

Our Priority

→ The priority is **understanding behavior**, not maximizing accuracy

OLS is natural for our goals

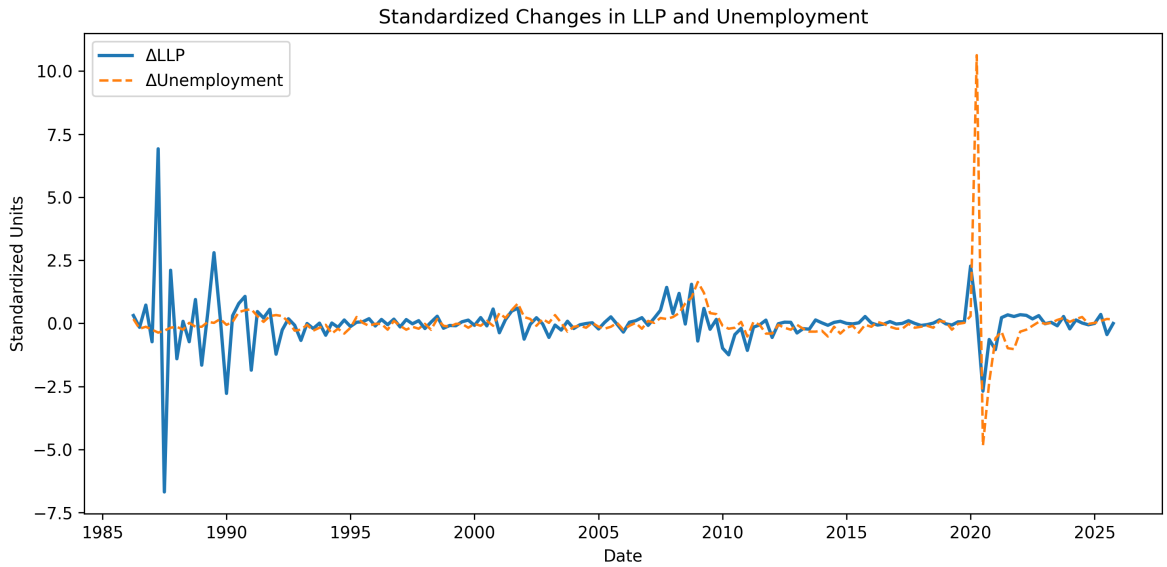
Results Overview

We build our analysis in stages:

1. Visual exploration
2. Single-variable models
3. Financial indicators
4. Timing structure (lags & leads)
5. Combined model
6. Regime analysis

→ Goal: Understand how LLP responds over time

Visual Analysis: LLP and Unemployment



- Standardized changes allow direct comparison
- Some periods show alignment
- Other periods show divergence
- Relationship is volatile, but the shape is suggestive of a pattern

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Baseline Model: Unemployment

Metric	Value
R^2	~0.05
Key Result	Weak relationship

- Some lagged effects are significant
- No consistent timing pattern
- LLP does not strongly track unemployment alone

First Insight

- Real economy variables provide **limited explanatory power**
- Suggests LLP is not driven by a single signal
- Need broader, more forward-looking indicators

Credit Spread Model

Metric	Value
R^2	~0.04
Key Result	Weak but significant contemporaneous effect

- Immediate response is significant
- Little persistence across lags

→ LLP reacts to financial stress, but weakly

Financial Conditions (NFCI)

Metric	Value
R^2	~0.23
Key Result	Stronger explanatory power

- Significant at lag 0 and lag 2
- Clearer dynamic pattern

→ Financial conditions better explain LLP behavior

Combining Signals

- The economy is **multi-dimensional**
- No single variable captures overall risk

In practice:

- Banks observe **many signals simultaneously**
- Risk emerges from a combination of factors
- A single-variable model may miss important information

Combined Model

Metric	Value
R^2	~0.45
Observations	154

- Substantial increase in explanatory power
- Multiple variables contribute simultaneously

→ LLP responds to a **combination of macro signals**

Full Model (All Variables)

Metric	Value
R^2	~0.59
Variables	30+

- Highest explanatory power observed
- Many coefficients insignificant
- Mixed signs across time

→ Model captures broad information, but lacks clarity

Interpreting the Full Model

- Many variables contain **overlapping information**
- Econometric pitfall of multicollinearity

Evidence:

- Loss of individual significance
- Instability across lags and leads
- Signals are **highly correlated**

Key Insight:

- LLP is influenced by a **complex web of signals**
- No single variable dominates, multivariate analysis outperforms univariate

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What Mattered Most

- Financial conditions (NFCI) remain important

- Unemployment still contributes
- Other variables add noise when combined

Refocusing the Model

- The full model includes many overlapping signals

To improve clarity:

- We focus on the strongest variables:
  - Financial Conditions (NFCI)
  - Unemployment
- Provides a clearer view of LLP behavior

Metric	Value

Core Model (NFCI + Unemployment)

Metric	Value
R^2	~0.45
Observations	~150

- Strong explanatory power
- More stable coefficients
- Captures the key drivers of LLP

One Last Adjustment:

Why Consider Regimes?

- Economic conditions are not constant over time

In particular:

- Crisis periods differ from normal periods
- Risk becomes more visible and urgent
- LLP behavior may change depending on the environment

Defining Regimes

We split the data into:

- **Normal periods**
- **Crisis periods** (e.g., 2008, COVID)

→ Allows us to test whether behavior changes under stress

Regime Analysis Method

- Extend the core model (NFCI + Unemployment)

Add interaction terms:

- Variable × Crisis indicator

→ Measures how responses differ during crisis periods

Regime Model Results

Metric	Value
R^2	~0.55

- Model fit improves further
- Several interaction terms are significant
- Evidence of different behavior in crisis periods

Crisis vs Normal Behavior

During crisis periods:

- Larger coefficient magnitudes
- More statistically significant effects
- Stronger and more persistent responses

→ LLP becomes more reactive under stress

Key Interpretation

- In normal times: → LLP adjusts gradually
- In crisis periods: → LLP responds more aggressively

→ Banks react more strongly when risk becomes clear

Circling back to the Research Question:

We now understand:

→ What LLP responds to
→ How responses change across environments

How did LLP respond over time?

Timing Framework

Our Core model was built to include:

- Leads (t+1, t+2) -> 1-2 quarters in advance
- Current values (t)
- Lags (t−1, t−2, t−3) -> 1-3 quarters of reaction time

→ Allows us to trace how LLP responds **over time**

Master Table of Results:

Timing Results (Normal vs Crisis)

Financial Conditions (NFCI)

Period	Lead 2	Lead 1	Lag 0	Lag 1	Lag 2	Lag 3
Normal	+	*	+	*	*	*
Crisis Effect	(−)	+	+	+	+	+

- + = significant positive
  − = significant negative
  ( ) = marginal significance ($p < 0.10$)

- * = not significant

Unemployment

Period	Lead 2	Lead 1	Lag 0	Lag 1	Lag 2	Lag 3
Normal	*	+	+	*	*	(−)
Crisis Effect	−	*	+	−	*	+

- = significant positive
  − = significant negative
  ( ) = marginal significance ($p < 0.10$)

- = not significant

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3 Important Takeaways:

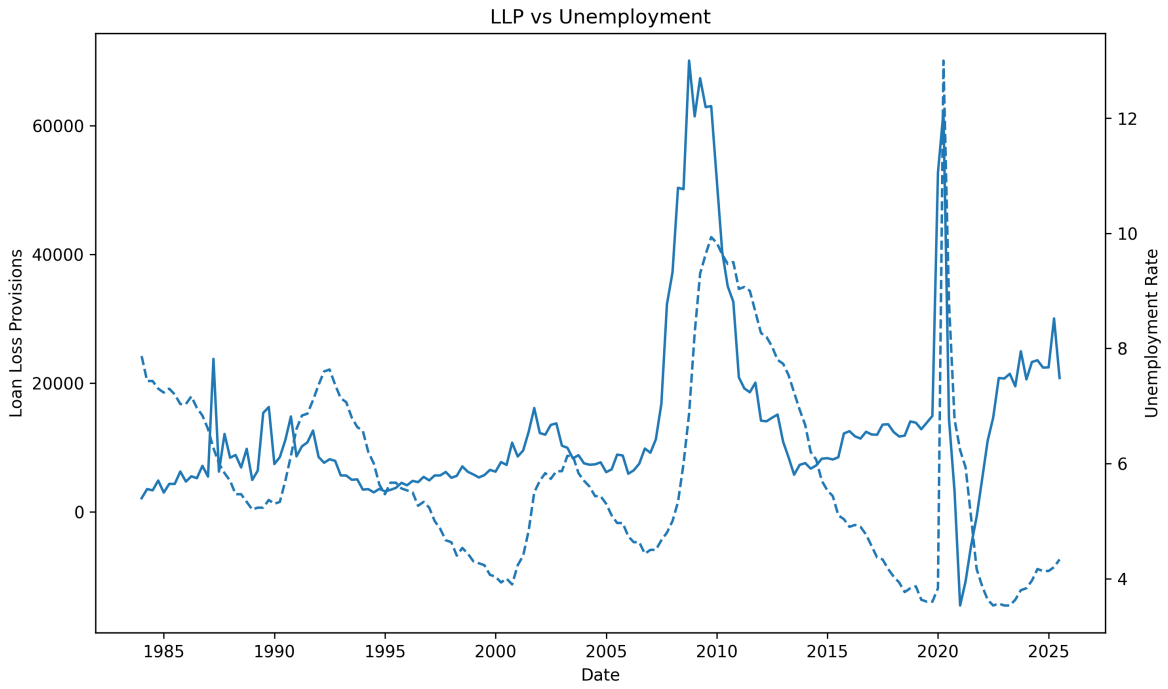
- Co-movement during crisis
- Diffuse reactions to economic shock
- Weak evidence of anticipation

Behavior Across Economic Conditions

- LLP more closely tracks macro conditions during crises
- In stable periods:
  - We observe weaker relationships
  - Less consistent adjustment patterns
  - Insignificant timing effects

→ Provisioning behavior varies across economic regimes

Visual Evidence



- Strong co-movement during crises (e.g., 2008)
- Contrary motion during stable periods
- Our lag/lead significance pattern is a reflection of this changing relationship

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Explanation:

- When risk is low, macro signals are less relevant to provisioning decisions
- During times of stability, banks may focus on other factors (e.g., competition, profitability, business metrics)
- Fear of being left behind - When the competition is not provisioning, it increases the opportunity cost of playing safe, even if risk is increasing in the background

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Diffuse Adjustment During Crises

- Economic shocks trigger immediate LLP responses
- Adjustments continue over multiple quarters
- Effects are distributed across time in a non-deterministic way

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→ LLP behaves as a **multi-period adjustment process**

Interpretation

- Banks **respond** to worsening conditions once they are forced to
- Adjustments are:
  - Gradual
  - Ongoing
  - Sometimes reversed
  - Not one-time decisions

→ Risk recognition evolves as information becomes clearer

Explanation:

- When risk becomes visible, banks adjust quickly to mitigate losses
- However, the adjustment process is not perfect or deterministic
- As conditions evolve, banks may need to adjust further, or even reverse course if conditions improve

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Weak Evidence of Anticipation

Evidence on Anticipation

- Some lead terms are statistically significant

However:

- No time-period or variable with two significant leads
- No time-period or variable with two leads of the same sign
- No stable pattern across variables

→ No strong evidence of systematic anticipation

Explanation:

- To make the claim that banks anticipate risk, we would need consistent evidence of significant lead effects across multiple variables and time periods
- The lack of significant leads, and the incosistent pattern of signs, suggests that banks do not systematically adjust LLP in anticipation of future economic changes

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Interpreting Lead Effects

- Macroeconomic conditions are persistent
- LLP responds over multiple quarters

My Theory

→ Lead significance may reflect:

- Ongoing adjustment to prior shocks
- Overlapping timing effects

→ Not true forward-looking behavior

Final Interpretation

- LLP **responds** to worsening macroeconomic conditions, especially when crisis is unambiguous and risk is visible
- Adjustments are:
  - Immediate
  - Gradual
  - Distributed over time
- Responses strengthen during shocks
- Little evidence of systematic anticipation

Answer to Research Question

Do banks anticipate macroeconomic risk?

- Limited evidence of anticipation
- Provisioning responds:
  - Contemporaneously
  - With lagged adjustments

In conclusion:

Provisioning behavior is **reactionary and state-dependent**

Role of Data Science

- Current behavior is largely reactionary

→ Opportunity to:

- Improve early risk detection
- Incorporate forward-looking signals
- Enhance consistency in provisioning decisions

Key Takeaway

→ Loan loss provisioning is **not anticipatory**

→ It is a **gradual, reactionary process**

→ Stronger during periods of economic stress